

Challenge 1

Data needed for this challenge: - athlete_events.csv - paintings.csv

Part 1. The Athlete Dataset

1. Read the dataset “athlete_events.csv”, and check the first few rows:

```
# Type your code here
```

2. **Data Description:** Please use the necessary commands and codes and briefly describe this data with a short writing paragraph answering the following questions.

```
# Type your code here
```

- (1) What is the dimension of the data (# of rows and columns)?
- (2) What do the rows and columns mean in this data?
- (3) What is the unit of observation? In other words, what does each case mean in this data?
- (4) According to the lecture, is this a “tidy” data?

3. **Data Transformation:** use necessary commands and codes and answer the following questions.

```
# Type your code here
```

- (1) How many unique sports, unique teams, and total unique names are in the data?
- (2) How many years does this data record?
- (3) Summarize the min, mean, median, and max of “Age” and “Weight”. (Must use summarize())
- (4) (Optional) Summarize the min, mean, median, and max of “Height” by decade.

Part 2. Paintings

1. Read the dataset “paintings.csv”.

```
# Type your code here
```

2. Data Description: Please use the necessary commands and codes and briefly describe this data with a short writing paragraph answering the following questions.

```
# Type your code here
```

- (1) What is the dimension of the data (# of rows and columns)?
- (2) What do the rows and columns mean in this data?
- (3) What is the unit of observation? In other words, what does each case mean in this data?
- (4) According to the lecture, is this a “tidy” data?

3. Data Transformation: use necessary commands and codes, and answer the following questions.

```
# Type your code here
```

- (1) How many unique museums are in the data?
- (2) What are the range and average of the following variable “year”? Must use summarize().
- (3) How many missing data (NA) are in the following variables: “year” and “depicts”? (tips: use is.na())
- (4) Which museum have the most paintings of Salvador Dalí?